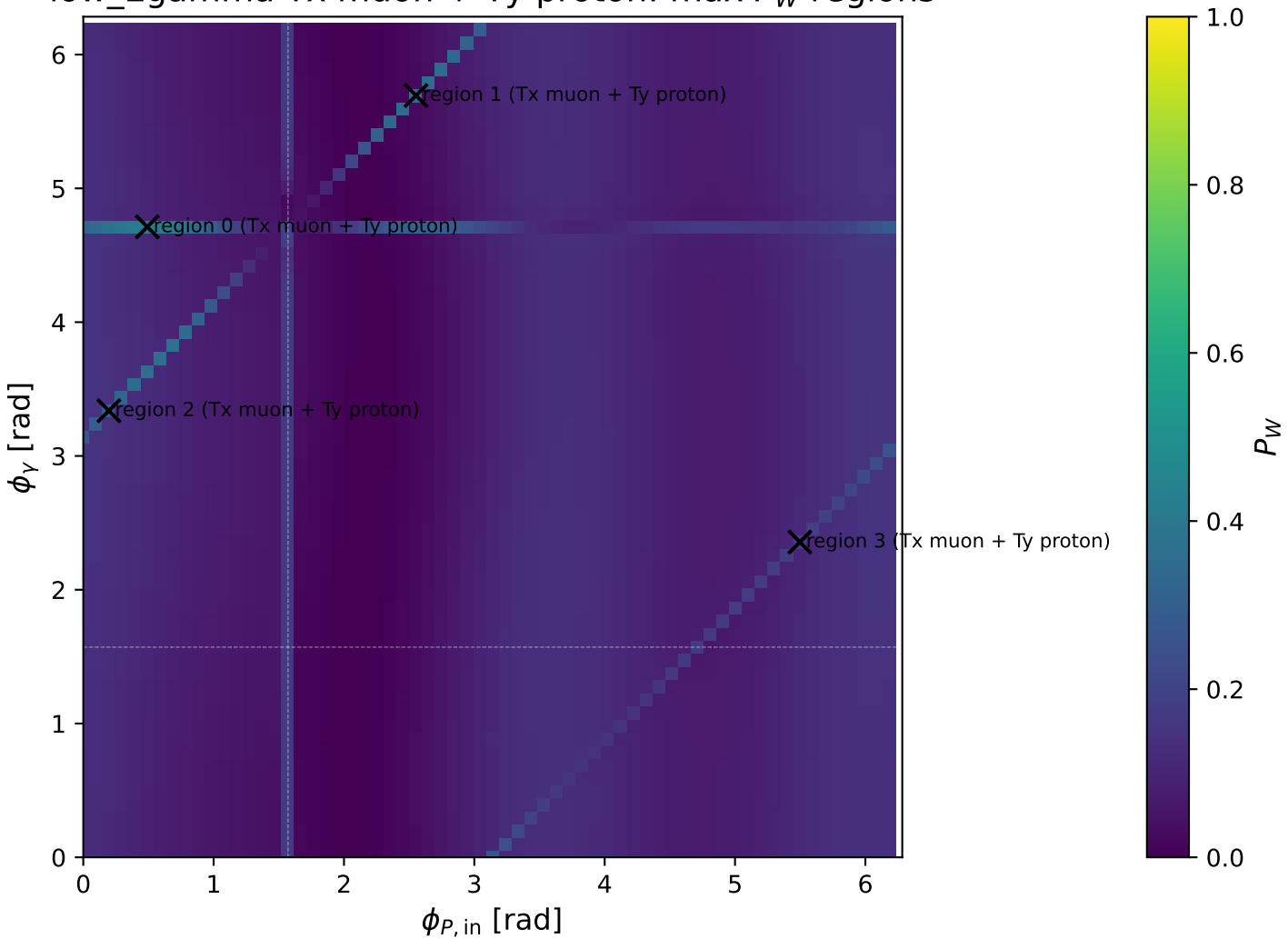
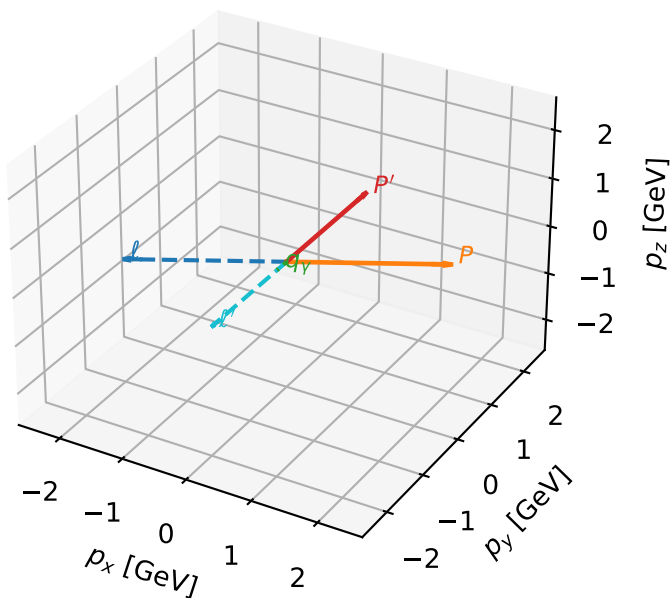


low\_Egamma Tx muon + Ty proton: max  $P_W$  regions



# Tx muon + Ty proton characteristic max $P_W$ configuration

3D momenta

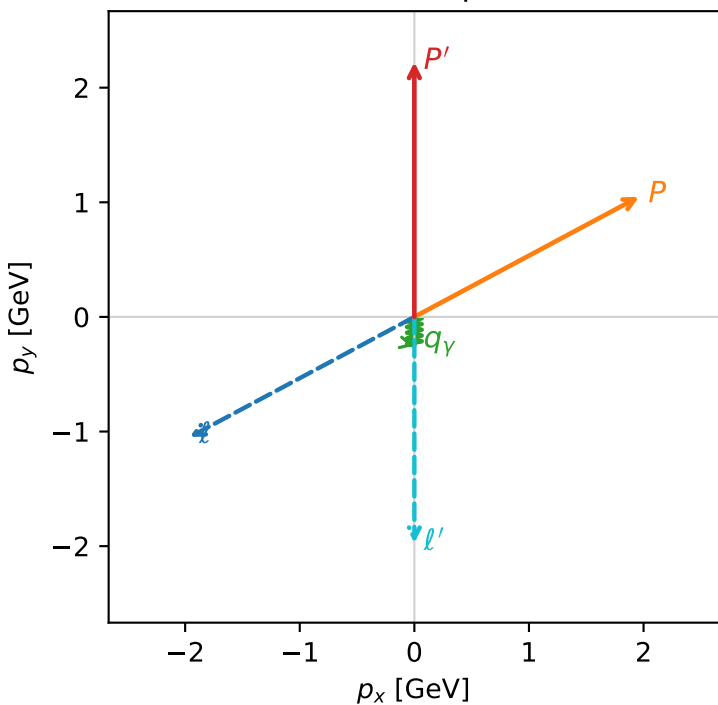


w\_purity\_low\_egamma\_txtty\_region\_0 P\_W=0.412937  
 region: low\_Egamma\_TxTy\_region\_0  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

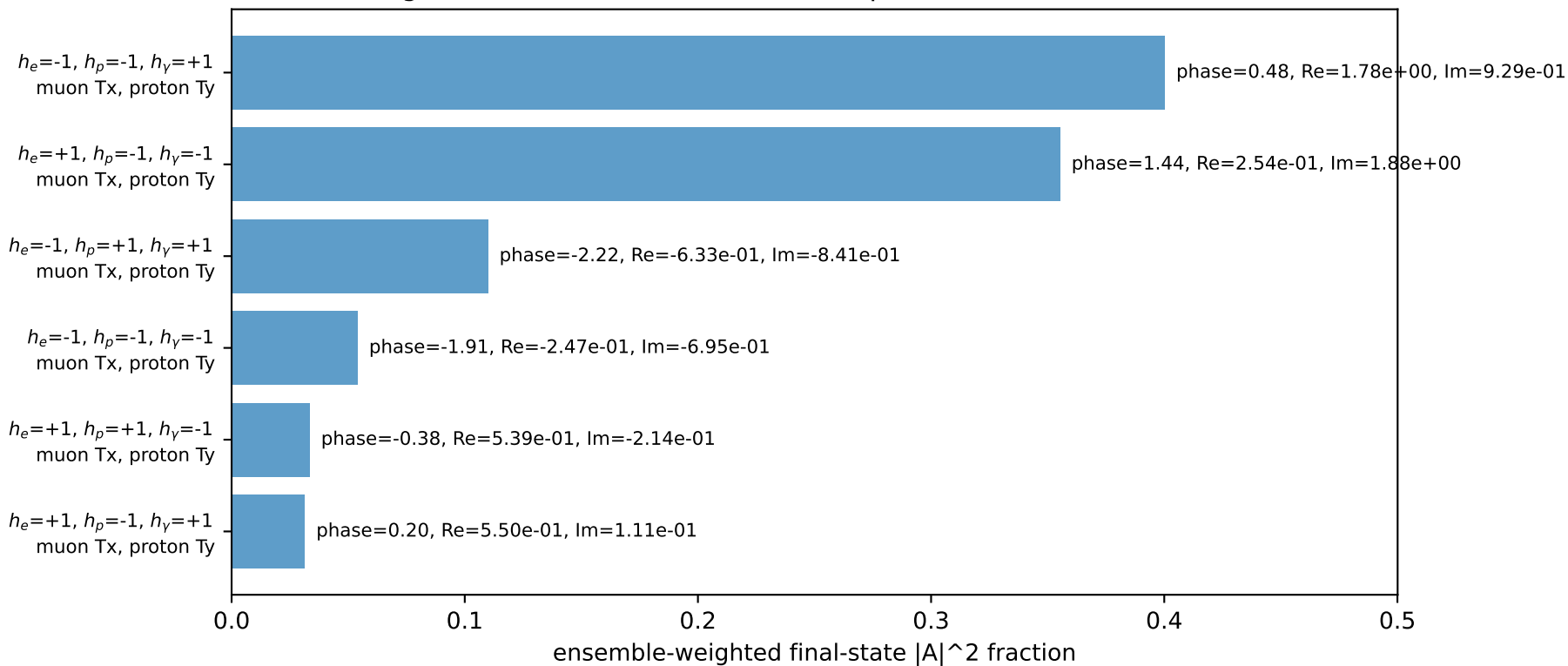
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=2.22295$   
 $\theta_{in}=1.57079$ ,  $\phi_l=3.63247$ ,  $\phi_P=0.490874$   
 $E_Y=0.25$ ,  $\phi_Y=4.71239$   
 $Q^2=4.63715$ ,  $x_B=0.666737$ ,  $t=-5.22456$ ,  $W^2=3.19769$ ,  $y=0.337215$   
 $\theta(l', \gamma)=0$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= -1.9606$   $p_y= -1.0480$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 1.9606$   $p_y= 1.0480$   $p_z= 0.0000$   
 $l'$   $E= 1.9758$   $p_x= 0.0000$   $p_y= -1.9729$   $p_z= 0.0000$   
 $P'$   $E= 2.4127$   $p_x= 0.0000$   $p_y= 2.2229$   $p_z= 0.0000$   
 $q_Y$   $E= 0.2500$   $p_x= -0.0000$   $p_y= -0.2500$   $p_z= 0.0000$

Transverse plane

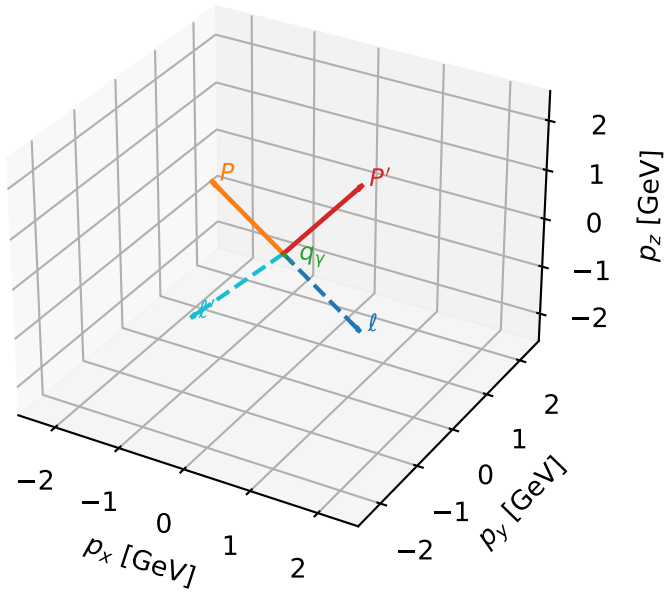


Leading initial-ensemble/final-state components (N=6, retained=98.3%)



# Tx muon + Ty proton characteristic max $P_W$ configuration

3D momenta

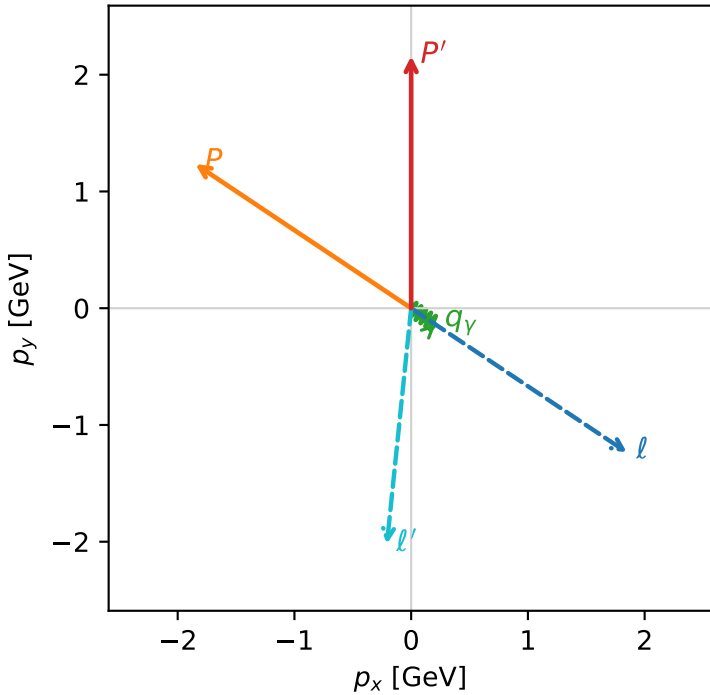


w\_purity\_low\_egamma\_txty\_region\_1  $P_W=0.355039$   
 region: low\_Egamma\_TxTy\_region\_1  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

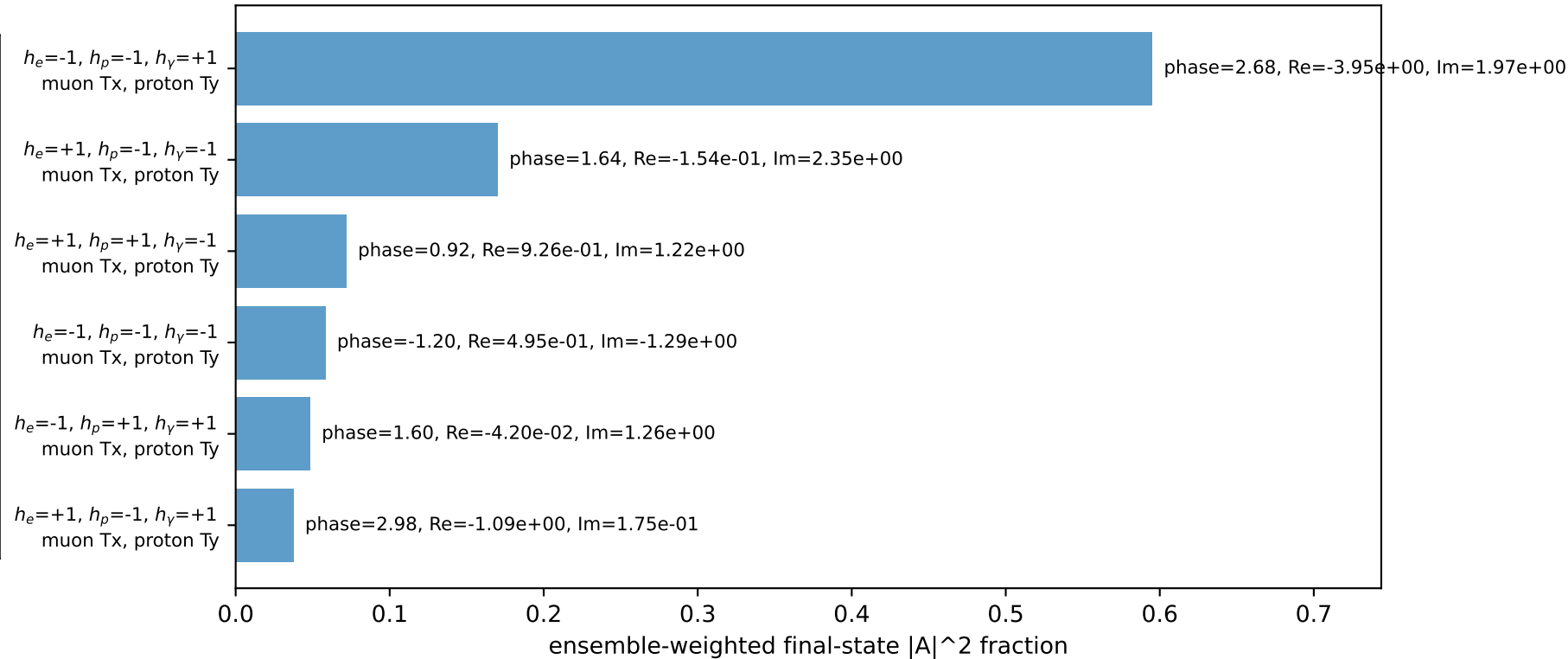
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=2.15954$   
 $\theta_{in}=1.57079$ ,  $\phi_l=5.69414$ ,  $\phi_p=2.55254$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=5.69414$   
 $Q^2=4.80884$ ,  $x_B=0.730166$ ,  $t=-4.26796$ ,  $W^2=2.65696$ ,  $y=0.319322$   
 $\theta(l', \gamma)=62.1234$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= 1.8484$   $p_y= -1.2351$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -1.8484$   $p_y= 1.2351$   $p_z= 0.0000$   
 $l'$   $E= 2.0341$   $p_x= -0.2079$   $p_y= -2.0207$   $p_z= 0.0000$   
 $P'$   $E= 2.3545$   $p_x= 0.0000$   $p_y= 2.1595$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.2079$   $p_y= -0.1389$   $p_z= 0.0000$

Transverse plane

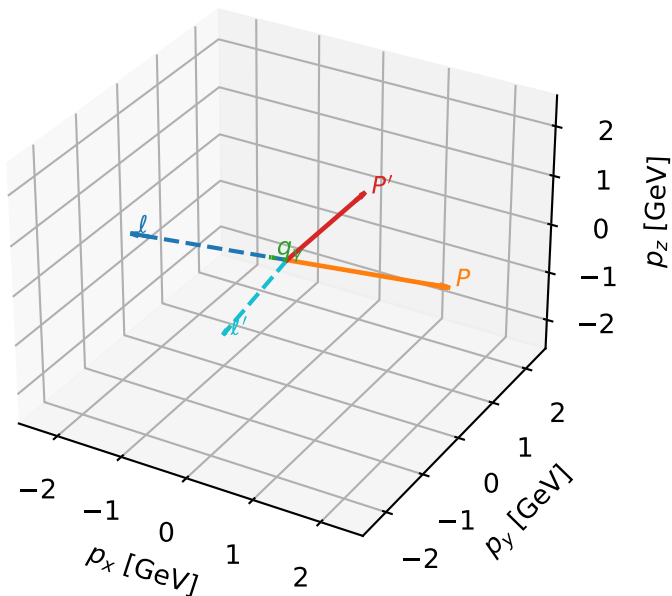


Leading initial-ensemble/final-state components (N=6, retained=98.1%)



# Tx muon + Ty proton characteristic max $P_W$ configuration

3D momenta

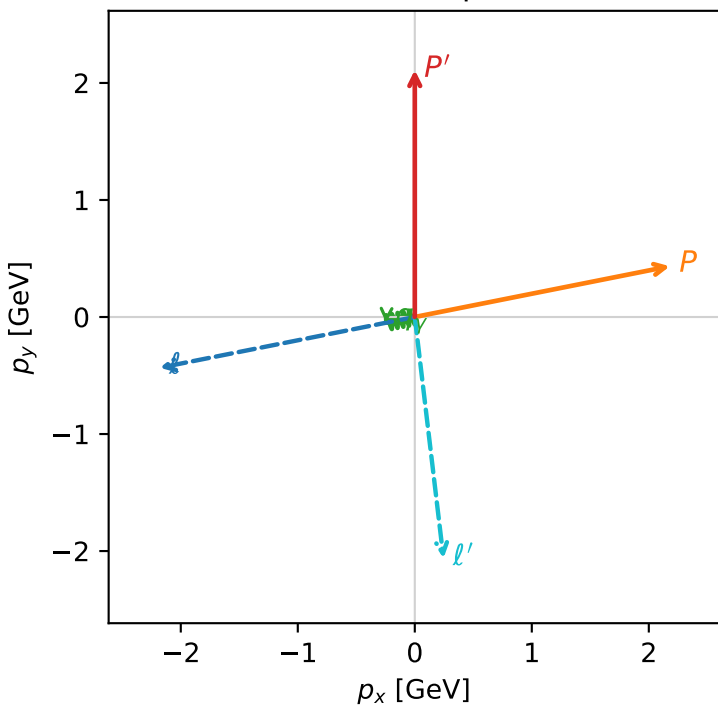


w\_purity\_low\_egamma\_txtty\_region\_2 P\_W=0.312443  
 region: low\_Egamma\_TxTy\_region\_2  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

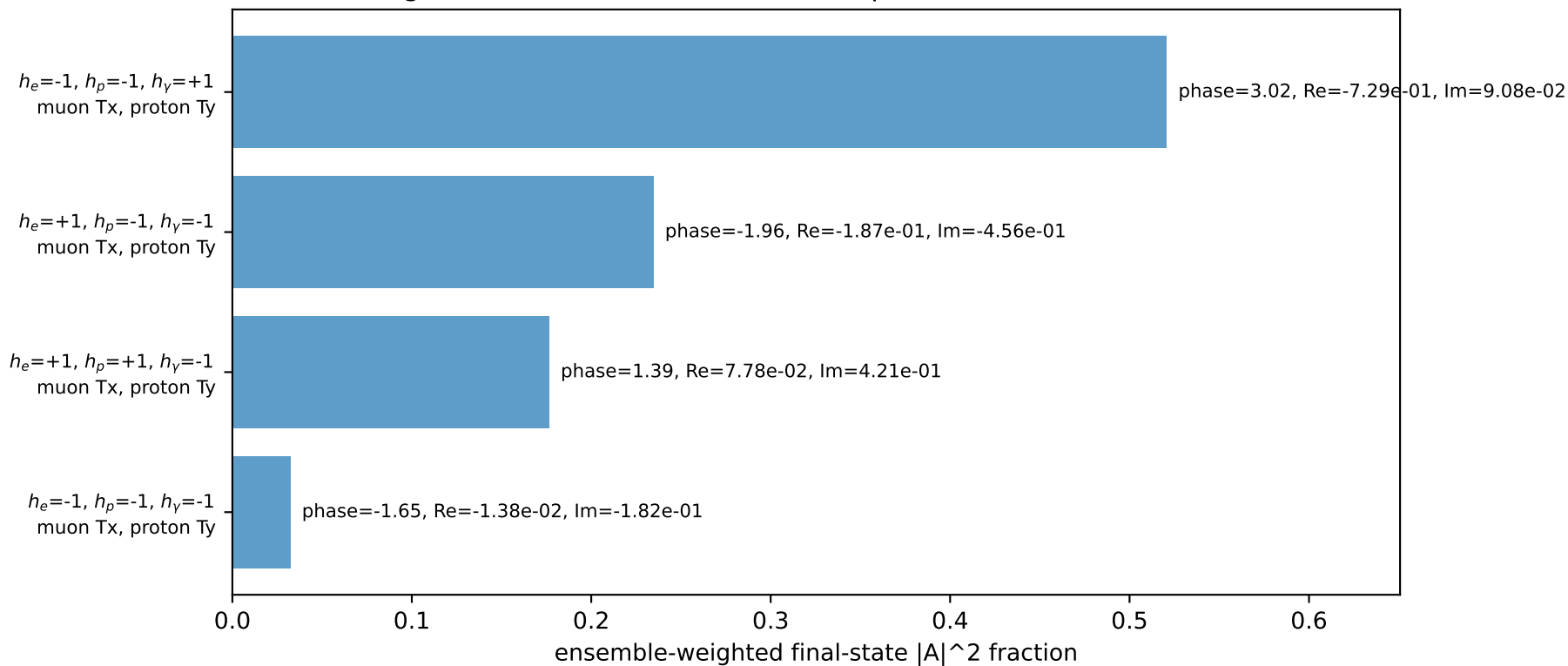
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=2.11051$   
 $\theta_{in}=1.57079$ ,  $\phi_l=3.33794$ ,  $\phi_P=0.19635$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=3.33794$   
 $Q^2=8.51246$ ,  $x_B=0.862186$ ,  $t=-7.55511$ ,  $W^2=2.2405$ ,  $y=0.4787$   
 $\theta(l', \gamma)=85.5322$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= -2.1804$   $p_y= -0.4337$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 2.1804$   $p_y= 0.4337$   $p_z= 0.0000$   
 $l'$   $E= 2.0790$   $p_x= 0.2452$   $p_y= -2.0617$   $p_z= 0.0000$   
 $P'$   $E= 2.3096$   $p_x= 0.0000$   $p_y= 2.1105$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= -0.2452$   $p_y= -0.0488$   $p_z= 0.0000$

Transverse plane

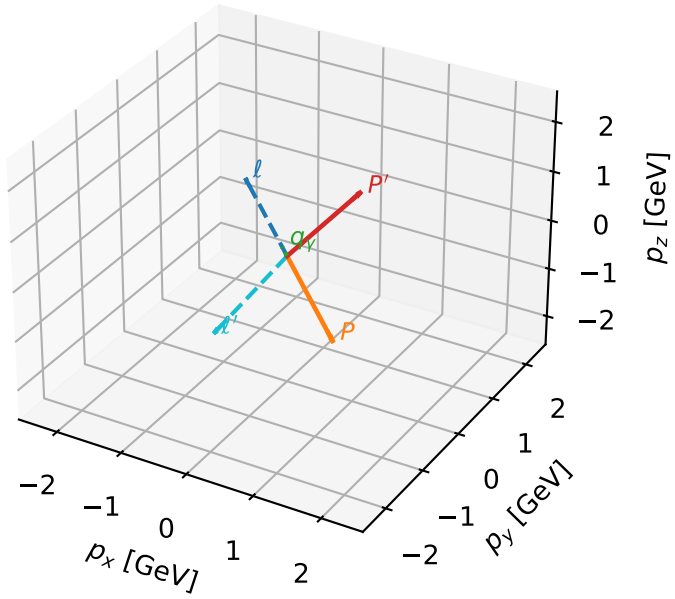


Leading initial-ensemble/final-state components (N=4, retained=96.4%)



# Tx muon + Ty proton characteristic max $P_W$ configuration

3D momenta

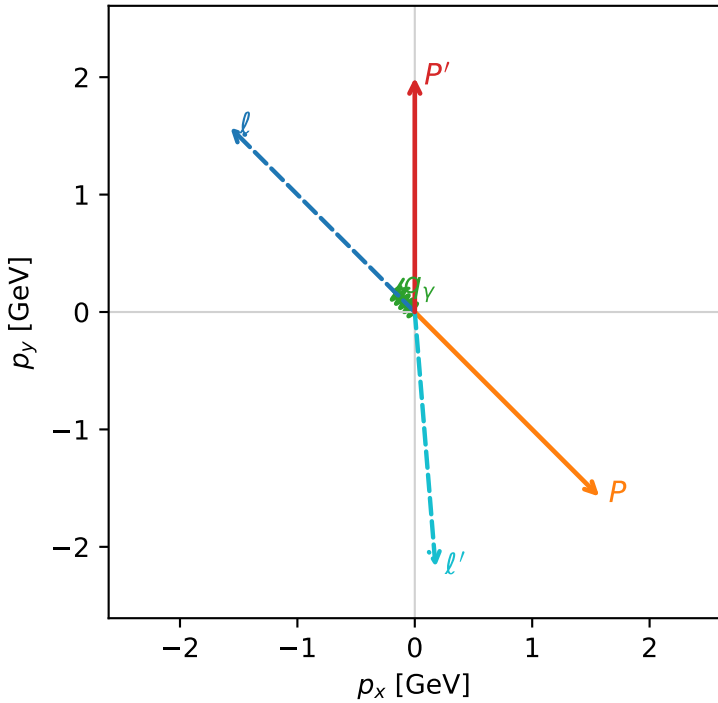


w\_purity\_low\_egamma\_txtty\_region\_3 P\_W=0.179824  
 region: low\_Egamma\_TxTy\_region\_3  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=1.99631$   
 $\theta_{in}=1.57079$ ,  $\phi_l=2.35619$ ,  $\phi_p=5.49779$   
 $E_Y=0.25$ ,  $\phi_Y=2.35619$   
 $Q^2=17.0818$ ,  $x_B=0.977284$ ,  $t=-15.1608$ ,  $W^2=1.27689$ ,  $y=0.847466$   
 $\theta(l', \gamma)=139.651$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= -1.5720$   $p_y= 1.5720$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 1.5720$   $p_y= -1.5720$   $p_z= 0.0000$   
 $l'$   $E= 2.1828$   $p_x= 0.1768$   $p_y= -2.1731$   $p_z= 0.0000$   
 $P'$   $E= 2.2057$   $p_x= 0.0000$   $p_y= 1.9963$   $p_z= 0.0000$   
 $q_Y$   $E= 0.2500$   $p_x= -0.1768$   $p_y= 0.1768$   $p_z= 0.0000$

Transverse plane



$h_e=-1$ ,  $h_p=-1$ ,  $h_Y=+1$   
 muon Tx, proton Ty

$h_e=+1$ ,  $h_p=+1$ ,  $h_Y=-1$   
 muon Tx, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=97.3%)

